

Microeconomic Analysis

Static Optimisation

HT 2025

Inés Moreno de Barreda

(ines.morenodebarreda@economics.ox.ac.uk)

February 17, 2025

Static Optimisation

Topics

- I Unconstrained Optimisation (Week 2)
- II Optimisation with Equality Constraints
- III Optimisation with Inequality Constraints
- IV Concave and Convex Problems
- V Envelope Theorems

General Optimisation Problem

The optimisation problem is to maximise (or minimise) a real-valued f over some subset S :

$$\text{maximise } f(\mathbf{x}), \text{ subject to } \mathbf{x} \in S$$

Throughout these lectures we will assume that f is C^2 on S .

Weierstrass Theorem: if S is a compact set, then f attains a global maximum and global minimum on S .

If S is not compact, the optimisation problem may not have a solution.

When searching for solutions of optimisation problems we need to consider two types of optimum:

- an **interior optimum**: occurs at an interior point of S ;
- a **constrained optimum**: occurs on the boundary of S , where the constraint binds.

Steps to find interior local optima

1. Solve the FOCs to find all the candidates for interior maxima. ($Df = \mathbf{0}$)
2. Check the SOC: points where the Hessian is negative definite are local maxima.
3. Evaluate the function at these points to find a possible global maximum.

But this procedure does not:

- tell you what happens at a point where the Hessian is negative semi-definite — it may or may not be a local maximum;
- tell you whether the interior local optimum where f is largest or smallest is actually a global optimum — if S is not compact the global optimum might not exist, and if S is not open the global optimum may be a constrained one.

II. Optimisation with Equality Constraints

Optimisation with Equality Constraints

Suppose $\mathbf{h} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a C^2 (vector-valued) function on $\mathbb{R}^n$.

$$\mathbf{h}(\mathbf{x}) = \begin{pmatrix} h_1(\mathbf{x}) \\ h_2(\mathbf{x}) \\ \vdots \\ h_m(\mathbf{x}) \end{pmatrix} \quad \text{and} \quad D\mathbf{h} = \begin{pmatrix} \frac{\partial h_1}{\partial x_1} & \cdots & \frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_m}{\partial x_1} & \cdots & \frac{\partial h_m}{\partial x_n} \end{pmatrix}$$

Consider the problem:

maximise $f(\mathbf{x})$ subject to $\mathbf{h}(\mathbf{x}) = \mathbf{a}$

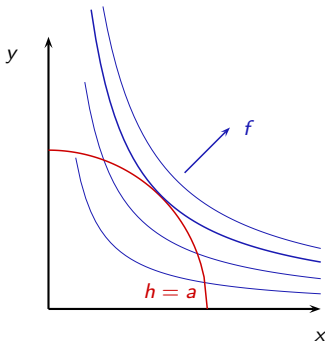
or equivalently:

$$\max f(x_1, \dots, x_n) \quad \text{s.t.}$$

$$\begin{cases} h_1(x_1, \dots, x_n) = a_1 \\ \dots \\ h_m(x_1, \dots, x_n) = a_m \end{cases}$$

Geometric Intuition

Consider the problem of maximising $f(x, y)$ subject to $h(x, y) = a$:

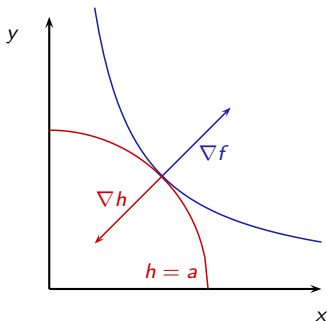


At an optimum it must not be possible to move in any direction of the constraint set which increases f .

Hence the level curve of f must be tangent to the curve $h = a$ at the optimum.

Geometric Intuition

Equivalently, if the level curves are tangent, the normal vectors to the curves must point in the same or opposite directions:



Hence

$$\nabla f = \lambda \nabla h$$

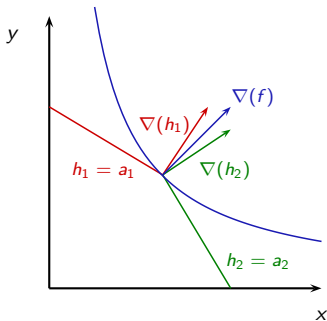
for some λ or equivalently

$$\frac{\partial f}{\partial x} = \lambda \frac{\partial h}{\partial x}, \quad \frac{\partial f}{\partial y} = \lambda \frac{\partial h}{\partial y}$$

Geometric Intuition

More generally, ∇f must be a linear combination of the gradients of the constraints at the optimal point: $\nabla f = \lambda_1 \nabla h_1 + \dots + \lambda_m \nabla h_m$.

In a picture if maximising $f(x, y)$ subject to $h_1(x, y) = a_1$ and $h_2(x, y) = a_2$:



The Lagrangian Approach

Define the Lagrangian $L : \mathbb{R}^{n+m} \rightarrow \mathbb{R}$ as:

$$L(\mathbf{x}; \boldsymbol{\lambda}) = f(\mathbf{x}) - \sum_{j=1}^m \lambda_j (h_j(\mathbf{x}) - a_j)$$

Let

$$D_{\mathbf{x}}L(\mathbf{x}; \boldsymbol{\lambda}) = Df(\mathbf{x}) - \sum_j \lambda_j Dh_j(\mathbf{x})$$

Necessary Conditions: Lagrange's Theorem

If $\mathbf{x}^*$ is a local maximum of f subject to the constraint $\mathbf{h}(\mathbf{x}) = \mathbf{a}$, and the matrix $D\mathbf{h}(\mathbf{x}^*)$ is of full rank, m , then there exists $\boldsymbol{\lambda}^* = (\lambda_1^*, \dots, \lambda_m^*)$ such that

$$D_{\mathbf{x}}L(\mathbf{x}^*; \boldsymbol{\lambda}^*) = 0$$

The Lagrangian Approach

The condition that the matrix $D\mathbf{h}(\mathbf{x}^*)$ must be of rank m is called the **constraint qualification**.

The Lagrange's theorem implies that (as long as the constraint qualification is satisfied) we can look for a solution to our problem by solving the following $n + m$ **First Order Conditions** for the Lagrangian:

$$\begin{aligned}\frac{\partial L}{\partial x_i}(\mathbf{x}^*; \boldsymbol{\lambda}^*) &= 0 & \text{for } i = 1, \dots, n \\ h_j(\mathbf{x}^*) &= a_j & \text{for } j = 1, \dots, m\end{aligned}$$

Interpretation of Lagrange Multipliers

As we will prove later, λ_j measures the sensitivity of the value of the objective function at $\mathbf{x}^*$ to a relaxation of the constraint $h_j(\mathbf{x}) = a_j$. That is,

$$\lambda_j = \frac{\partial f(\mathbf{x}^*(\mathbf{a}))}{\partial a_j}$$

The Lagrangian Approach

Proof of Lagrange's Theorem: (for $n = 2, m = 1$)

Consider the problem of maximising $f(x_1, x_2)$ subject to $h(x_1, x_2) = 0$. We want to show that there is a λ satisfying the Lagrangian FOC at the (local) optimum $\mathbf{x}^*$

$$f_{x_1}(\mathbf{x}^*) - \lambda h_{x_1}(\mathbf{x}^*) = 0 \quad (1)$$

$$f_{x_2}(\mathbf{x}^*) - \lambda h_{x_2}(\mathbf{x}^*) = 0 \quad (2)$$

The **constraint qualification** implies that at least one of $h_{x_1}(\mathbf{x}^*)$ and $h_{x_2}(\mathbf{x}^*)$ is non-zero. Suppose $h_{x_2}(\mathbf{x}^*) \neq 0$.

By the implicit function theorem, there exists a differentiable function g , with $x_2^* = g(x_1^*)$ and such that $h(x_1, g(x_1)) = 0$ in a neighbourhood of x_1^* . Moreover $g'(x_1^*) = -h_{x_1}(\mathbf{x}^*)/h_{x_2}(\mathbf{x}^*)$.

Since $\mathbf{x}^*$ is a local optimum, x_1^* must maximise $f(x_1, g(x_1))$ locally. Hence

$$0 = f_{x_1}(\mathbf{x}^*) + f_{x_2}(\mathbf{x}^*)g'(x_1^*) = f_{x_1}(\mathbf{x}^*) - f_{x_2}(\mathbf{x}^*) \frac{h_{x_1}(\mathbf{x}^*)}{h_{x_2}(\mathbf{x}^*)} \quad (3)$$

Let $\lambda = f_{x_2}(\mathbf{x}^*)/h_{x_2}(\mathbf{x}^*)$. (??) is satisfied by definition. (??) is satisfied on account of (??).

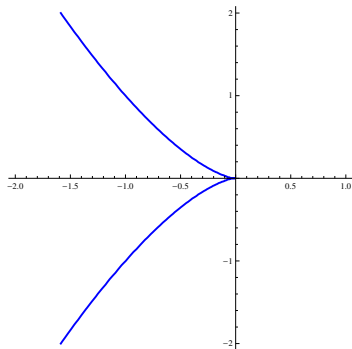
The Lagrangian Approach

If the constraint qualification fails... we cannot apply the implicit function theorem! The Lagrangian approach might fail.

Consider

$$\max x \quad \text{subject to} \quad y^2 + x^3 = 0$$

The feasible region looks like:



The optimum is clearly $x = 0, y = 0$.

$$Df = (1, 0) \text{ and } Dh = (3x^2, 2y).$$

So $Df(0, 0) \neq \lambda Dh(0, 0)$ for any λ !

Note that $Dh(0, 0) = (0, 0)$, so

$$\text{Rank}(Dh(0, 0)) = 0 \neq 1$$

the constraint qualification fails.

The Lagrangian Approach

Let

$$D_{\mathbf{x}}^2 L(\mathbf{x}; \boldsymbol{\lambda}) = D^2 f(\mathbf{x}) - \sum_j \lambda_j D^2 h_j(\mathbf{x})$$

Second Order sufficient conditions

Suppose that $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$ satisfies the constraint qualification and the first order conditions. Then:

- (a) if $\mathbf{v}^T D_{\mathbf{x}}^2 L(\mathbf{x}^*; \boldsymbol{\lambda}^*) \mathbf{v} < 0$ for all $\mathbf{v} \neq \mathbf{0}$ in $\mathbb{R}^n$ for which $D\mathbf{h}(\mathbf{x}^*)\mathbf{v} = 0$ then $\mathbf{x}^*$ is a strict local maximum.
- (b) if $\mathbf{v}^T D_{\mathbf{x}}^2 L(\mathbf{x}^*; \boldsymbol{\lambda}^*) \mathbf{v} > 0$ for all $\mathbf{v} \neq \mathbf{0}$ in $\mathbb{R}^n$ for which $D\mathbf{h}(\mathbf{x}^*)\mathbf{v} = 0$ then $\mathbf{x}^*$ is a strict local minimum.

The intuition for this condition comes from a Taylor expansion of $L(\mathbf{x}; \boldsymbol{\lambda}^*)$ around $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$, along the constraints.

Intuition for Second Order Conditions

How do we check for the second order conditions?

Given A ($n \times n$ matrix) and B ($m \times n$ matrix of full rank m where m is the number of binding constraints), we need to determine whether:

$$\begin{aligned} &\mathbf{v}^T A \mathbf{v} > 0 \text{ for all } \mathbf{v} \neq \mathbf{0} \text{ in } \mathbb{R}^n \text{ where } B\mathbf{v} = \mathbf{0} \\ &\text{or } \mathbf{v}^T A \mathbf{v} < 0 \text{ for all } \mathbf{v} \neq \mathbf{0} \text{ in } \mathbb{R}^n \text{ where } B\mathbf{v} = \mathbf{0} \end{aligned}$$

Example:

Consider the quadratic form:

$$Q(x, y) = ax^2 + 2bxy + cy^2 = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

and the linear constraint:

$$\alpha x + \beta y = 0$$

with either α or β non zero (CQ).

Intuition for Second Order Conditions

Suppose $\alpha \neq 0$. We can then write $x = -\frac{\beta y}{\alpha}$ and substitute in the quadratic form:

$$Q\left(-\frac{\beta y}{\alpha}, y\right) = a\left(-\frac{\beta y}{\alpha}\right)^2 + 2b\left(-\frac{\beta y}{\alpha}\right)y + cy^2 = \frac{a\beta^2 - 2b\alpha\beta + c\alpha^2}{\alpha^2}y^2$$

Therefore, Q is definite positive (negative) over the linear constraint if and only if

$$a\beta^2 - 2b\alpha\beta + c\alpha^2 > (<)0$$

But note that:

$$a\beta^2 - 2b\alpha\beta + c\alpha^2 = -\det \begin{pmatrix} 0 & \alpha & \beta \\ \alpha & a & b \\ \beta & b & c \end{pmatrix}$$

So we need to check for the determinant of the **bordered** matrix.

In general, there will be n variables and m constraints, so there will be $n - m$ degree of freedoms (directions to move) and hence $n - m$ conditions to check.

The Bordered Hessian

Let C be the square matrix of order $m + n$ formed by *bordering* the matrix A by the matrix B :

$$C = \begin{pmatrix} \mathbf{0} & B \\ B^T & A \end{pmatrix},$$

and let $|C_r|$ denote the leading principal minor that ends with a_{rr} , so that each $|C_r|$ is the determinant of a square matrix of order $m + r$.

Bordered Hessian Test

- 1 $\mathbf{v}^T A \mathbf{v} > 0$ for all $\mathbf{v} \neq \mathbf{0}$ in $\mathbb{R}^n$ where $B\mathbf{v} = \mathbf{0}$ iff $(-1)^m |C_r| > 0$ for $r = m + 1, \dots, n$
–the last $n - m$ leading principal minors have the same sign, namely $(-1)^m$.
- 2 $\mathbf{v}^T A \mathbf{v} < 0$ for all $\mathbf{v} \neq \mathbf{0}$ in $\mathbb{R}^n$ where $B\mathbf{v} = \mathbf{0}$ iff $(-1)^r |C_r| > 0$ for $r = m + 1, \dots, n$
–the last $n - m$ leading principal minors alternate in sign ending with $(-1)^n$

Second Order Conditions

Therefore, applied to our Lagrangian:

Construct the bordered Hessian:

$$\begin{pmatrix} \mathbf{0} & D\mathbf{h}(\mathbf{x}^*) \\ D\mathbf{h}(\mathbf{x}^*)^T & D_{\mathbf{x}}^2 L(\mathbf{x}^*; \boldsymbol{\lambda}^*) \end{pmatrix}$$

And check the sign of the last $n - m$ leading principal minors.

- If they alternate, ending with $(-1)^n$, then $\mathbf{x}^*$ is a **strict local maximum**.
- If they all have the same sign as $(-1)^m$, then $\mathbf{x}^*$ is a **strict local minimum**.

Steps to solve an equality-constrained maximisation problem

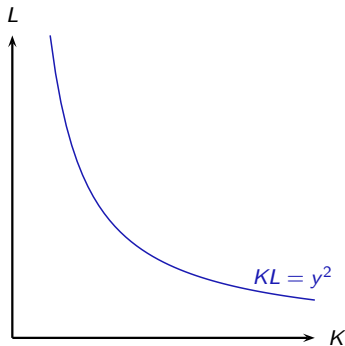
- 1 Start by considering whether the required global maximum exists.
- 2 Write down the matrix $D\mathbf{h}(x)$, and check whether there are any points where it does not have full rank – if so, these are candidates for maximum points.
- 3 Write down the Lagrangian and obtain the first order conditions. Solve to obtain the other candidate maxima.
- 4 If you already know that the global maximum exists, it is one of the candidates you have found. If you evaluate f at each of these points, you can identify it immediately.
- 5 Otherwise, write down the bordered Hessian, and check the second order conditions for each point satisfying the first order conditions. Then reconsider the global maximum.

An Example

$$\min_{K,L} rK + wL, \quad K^{1/2}L^{1/2} = y > 0 \quad (r > 0, w > 0 \text{ given})$$

Clearly $K > 0$ and $L > 0$; rewrite the constraint as $KL = y^2$.

- 1 The constraint set is not compact... Cannot call the Weierstrass theorem.



An Example

$$\min_{K,L} rK + wL, \quad KL = y^2 > 0 \quad (r > 0, w > 0 \text{ given})$$

2 $Dh = (L \quad K) \neq \mathbf{0}$, the constraint qualification cannot fail.

3 Lagrangian:

$$\mathcal{L}(K, L; \lambda) = rK + wL - \lambda(KL - y^2)$$

FOCs:

$$\text{FOC}_K : \quad r - \lambda L = 0$$

$$\text{FOC}_L : \quad w - \lambda K = 0$$

$$\text{Cs} : \quad KL = y^2$$

$$\text{FOCs} \implies \lambda = r/L = w/K > 0, \quad \text{so} \quad y^2 = K^2 r/w = L^2 w/r,$$

$$\text{i.e.} \quad K^* = y(w/r)^{1/2}, \quad L^* = y(r/w)^{1/2}$$

An Example

4 SOC

$$\text{Bordered Hessian } H = \begin{pmatrix} 0 & L & K \\ L & 0 & -\lambda \\ K & -\lambda & 0 \end{pmatrix}$$

$n = 2, m = 1, n - m = 1$, so need to check the sign of $|H|$ only.

$$|H| = -L(0 + \lambda K) + K(-\lambda L - 0) = -2\lambda KL < 0,$$

Hence the sign of $|H|$ coincides with sign of $(-1)^m$, *local* minimum.

Let $c^*(y)$ be the cost of using K^*, L^* :

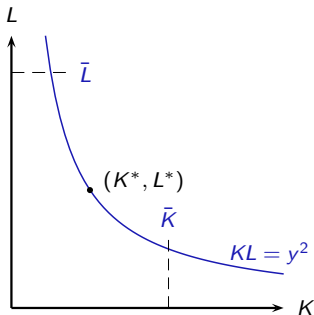
$$c^* = rK^* + wL^* = 2y(rw)^{1/2}.$$

An Example

5 Reconsider global minimum:

Define $\bar{K}$ & $\bar{L}$ by $r\bar{K} = c^*$, $w\bar{L} = c^*$.

Using $K \geq \bar{K}$ or $L \geq \bar{L}$ to produce y would cost more than $c^*(y)$, and so could not be optimal.



Add the additional constraints: $K \leq \bar{K}$ & $L \leq \bar{L}$.

Now, the constraint set is compact. But these constraints don't bind, so we get same unique solution to FOCs. **Global Minimum.**

Summing up:

- Given an optimisation problem with equality constraints, if the **constraint qualification** holds we have come up with **necessary conditions** for optimisation:
 - ▶ an optimiser needs to be a critical point of the Lagrangian.
- For those critical point we have provided sufficient conditions for optimality:
 - ▶ which we can check using the **Bordered Hessian**
- Points at which the constraint qualification are violated are also candidates for optima.
- And of course, global optima might not exist...